

[17]

SARDAR PATEL UNIVERSITY
M.Sc. IT Integrated (NC) Examination
Semester II (CBCS)

Thursday, Date: 17/09/2020

Session : Morning

Time : 10:00 am to 12:00 pm

Course Code: PS02CHIT03

Course Title : Advanced C and Introduction to Data Structures

Total Marks: 70

Q1. Multiple Choice Questions.(Attempt All)

[10]

1. Given the following declarations:
int a; double d; int *p; double *q;
Which of the following expression is allowed?
A. p=&a; C. q=&a;
B. d=&a; D. p=q;
2. malloc() function sets the initial value in memory as _____.
A. Zero C. One
B. Garbage D. Minus One
3. What is a proper mode of opening a file prg1.bin for writing file as a binary file?
A. r C. w
B. wb D. rb
4. _____ function is used to check end of file pointer.
A. eof() C. feof()
B. fileend() D. fileeof()
5. Which of the following allows a portion of memory to be shared by different types of data?
A. Array C. Union
B. Pointer D. Structure
6. A variable P is called pointer if ?
A. P contain the value of element DATA C. P contains the address of an element DATA
B. P contain the value and address both of element DATA D. P can store only constant memory address and it can not be changed
7. A data structure in which insertion and deletion of an elements occurs at only one end is known as _____.
A. Stack C. Queue
B. Linked List D. Array
8. Which of the following is TRUE for a Queue?
A. Its nature is LIFO C. Element is deleted from FRONT of the queue
B. It is a non-primitive data structure D. All of the above
9. A linked list is ____ type of data structure?
A. Linear C. Both a and b
B. Non-Linear D. None of these
10. A data structure in which insertion and deletion of an elements occurs at both the end is known as _____.
A. Priority queue C. Deque
B. Circular linked List D. Singly linked List

- Q2. Fill in the blanks and True/False. [One mark each]** [08]
1. _____ function is used to release the allocated memory using dynamic memory allocation functions.
 2. _____ data structure allows a portion of memory to be shared by different types of data.
 3. An operation that is used to insert an element from a stack is known as _____.
 4. A data structure that contains not only a data field but also contains pointer field is known as _____.
 5. & operator is used to declare a pointer variable. (True/False)
 6. Dynamic memory allocation is also known as run time or execution time memory allocation. (True/False)
 7. Element is deleted from BOTTOM of stack. (True/False)
 8. A linked list in which last node pointing to the first node is known as Circular Linked List. (True/False)
- Q3. Answer the following short questions (Attempt any TEN)** [20]
1. Explain '*' operator in brief.
 2. Explain malloc() function in brief with example.
 3. Give example to display concept of structure.
 4. Explain typedef in brief.
 5. List file modes available to manage the file in C.
 6. What is use of FILE data structure?
 7. List primitive data structures.
 8. List applications of STACK.
 9. List all operations of STACK and give representation of STACK.
 10. What is Linked List?
 11. List out various types of Linked List.
 12. State applications of Queue data structure.
- Q4. Answer the following long questions (Attempt any FOUR)** [32]
1. What is pointer ? Explain pointer declaration and initialization in detail with example.
 2. Explain concept of structure declaration and initialization with example.
 3. Differentiate between structure and union. Give one example of union.
 4. Explain concept of File Handling.
Also explain functions i) fopen() ii) fgets() iii) fclose()
 5. Explain a STACK with an example. Write any two algorithms for STACK operations.
 6. Write down advantages of data structure and Explain operations that can be performed on data structures in details.
 7. Explain Linked list in brief and write an algorithm to insert an element at the end of a Singly linked list.
 8. Explain Queue in brief and Write an algorithm to delete an element into a simple queue.